

Edward Lau

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Decision science professional with 3+ years of experience in the FinTech payments space, specializing in fraud and repayment risk. Skilled in Python + SQL, rule development, and AI orchestration. Proven track-record of protecting revenue by translating data into actionable strategies that neutralize emerging threats and mitigate financial losses.

Experience

BILL

San Jose, CA

Fraud Strategy Data Scientist

Feb 2026 – Present

- Reclaimed 15+ hours of weekly team capacity by engineering an automated AWS SageMaker/S3 metric pipeline with Python and SQL to eliminate coverage gaps and ensure fraud oversight across weekends and holidays
- Unlocked \$45,000+ in annualized revenue (9% lift) for a new product line by tuning risk controls and back testing threshold optimizations with Optuna to simultaneously maximize approval rates and fraud capture
- Informed proactive strategy shifts by developing an n8n and LLM-powered intelligence pipeline that synthesizes emerging threats into executive briefings and recommends actionable mitigations
- Collaborated with Product and Engineering to migrate analytics logic into a real-time Databricks Feature Store

Sezzle

Remote

Fraud Risk Data Analyst

Oct 2024 – Feb 2026

- Reduced fraud detection time by 90%+ and monthly losses by 20%+ by replacing reactive reporting with 10+ real-time SQL/dbt alerts that freeze high-risk accounts and instantly flag transaction anomalies
- Accelerated fraud investigation cycles by 2 hours per case by developing interactive Python scripts that automate SQL data retrieval and generate data visualizations of aggregated user trends
- Delivered \$30,000+ in cost savings and optimized vendor infrastructure by evaluating external fraud prevention services to recommend top-performing tools and decommission low-value features
- Increased fraud capture rate by implementing 4+ production rules to mitigate emerging risk typologies

Fraud Operations Analyst

Oct 2023 – Oct 2024

- Provided crucial data necessary for 13 risk mitigation methods by documenting and presenting data of confirmed fraud (account takeover, stolen cards, card compromise, first-party) from investigations
- Classified 20,000+ users by evaluating credit and fraud risks, providing training data for machine learning models
- Reduced financial losses by \$26,000+ by winning customer disputes and avoiding chargeback fees
- Lowered average fraud inquiry resolution time by 15%+ by interviewing, training, and mentoring 7 new hires

Fraud Analyst Intern

Oct 2022 – Oct 2023

- Received top performance rating in 2 months by solving 50+ tickets daily, reducing queue volume by 70%+ YoY
- Identified a data anomaly using SQL that impacted 40,000+ shoppers, enabling prompt incident resolution

Braathe Enterprises

Saratoga Springs, NY

Finance Intern

May 2021 – Sep 2021

- Promoted to a finance team co-lead within 1 month to co-manage new project ideas and 12 team members
- Analyzed 30+ investment securities and presented reports to provide recommendations for company portfolio

Skills

Languages: SQL, Python, R | **AI & Automation:** n8n, Zapier, Claude, Glean, NotebookLM

Data Visualization: Tableau, Microsoft Power BI, Redash, Looker Studio

Libraries (Python): pandas, NumPy, Matplotlib/Seaborn, scikit-learn, TensorFlow, Keras

Other: dbt, Excel (Lookups, Pivot Tables, Power Query, VBA), Accertify (Interceptas), Stripe

Activities

Freelance | Excel Business Data Analyst

Jun 2022 – Present

- Save 3 hours per sheet by processing monthly Amazon & Shopify data using Python, VBA, and Power Query
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Education

University of Southern California

May 2023

Bachelor of Science in Business Administration (Finance Emphasis) – Magna Cum Laude | **Minor: Applied Analytics**

Developed an unsupervised Isolation Forest framework to differentiate first-party misuse (friendly fraud) from third-party identity theft; analyzed multi-dimensional tree path lengths to isolate stealth behavioral anomalies with [X]% precision.

Benchmarked enterprise-grade anomaly detection frameworks (including GBM, Isolation Forest, and LOF detectors) to isolate first-party misuse, selecting the optimal gradient-boosted architecture to maximize friendly fraud detection precision.